

Data-Driven Analysis of Ferry Capacity Utilization and Operational Efficiency Using Python and Interactive Dashboards

Vikas Dharwan

Data Analytics Internship Project

Ferry Capacity Utilization & Operational Efficiency Analytics System

Abstract

This project presents a data analytics approach for evaluating ferry activity, capacity utilization, congestion, and idle-capacity patterns through time-based operational data. The analysis was performed using Python, Pandas, Jupyter Notebook, Plotly, and Streamlit. The dataset was transformed into analytical features such as Total Activity Load, Redemption Pressure Ratio, Operational Load Index (OLI), Idle Capacity, Time Band, Season, and Congestion indicators. Exploratory analysis was used to identify high-activity periods, operational hotspots, and differences between weekdays and weekends. An interactive Streamlit dashboard was developed to present KPIs, visualizations, filtered data, data-quality checks, and operational recommendations. The project demonstrates how operational data can be converted into actionable insights for capacity planning and resource allocation.

Keywords: Data Analytics; Ferry Operations; Capacity Utilization; Python; Pandas; Streamlit; Plotly; Operational Efficiency

1. Introduction

Transportation systems generate large volumes of operational data that can be used to understand demand, capacity utilization, congestion, and periods of under-utilization. Ferry services are particularly suitable for time-based analytics because passenger activity can vary substantially by hour, day type, season, and other operational conditions.

The objective of this project was to develop a practical data analytics system for examining ferry activity and operational efficiency. Rather than relying only on raw transaction counts, the project engineered analytical measures that describe the relationship between sales activity, redemptions, and operational load. The final solution combines exploratory data analysis with an interactive dashboard so that analytical findings can be communicated in a form that is easier for operational users to interpret.

2. Problem Statement

Ferry operators need to understand when demand is high, when capacity may be under pressure, and when available

capacity is likely to remain unused. Raw sales and redemption records do not directly provide a complete operational

picture. A structured analytics workflow is therefore required to transform timestamped activity records into meaningful indicators and visual summaries.

The problem addressed by this project is the design of a data analytics system that can identify temporal activity

patterns, congestion hotspots, idle-capacity periods, and differences across seasons and day types, while providing

an interactive interface for exploring the results.

3. Objectives

The main objectives were:

- Clean and prepare the ferry activity dataset for analysis.
- Create time-based and operational features from the source records.
- Measure overall activity, operational load, congestion, and idle-capacity patterns.
- Compare activity across hours, seasons, weekdays, weekends, and days of the week.
- Identify operational hotspots that may require attention.
- Develop an interactive Streamlit dashboard for KPI monitoring and exploration.
- Provide data-driven recommendations that can support capacity planning and operational decision-making.

4. Dataset and Feature Engineering

The project used timestamped ferry activity data containing fields such as Timestamp, Sales Count, and Redemption

Count. The timestamp was converted into a datetime representation and used to derive Year, Month, Day, DayOfWeek,

Hour, DayType, Time Band, and Season.

Additional analytical variables were created. Total Activity Load represents combined activity from sales and redemptions. Redemption Pressure Ratio describes redemption activity relative to sales activity. The Operational Load Index (OLI) was used as an operational indicator, while Low Activity, Idle Capacity, and Congestion were used

to classify periods according to operational conditions.

These engineered variables allowed the analysis to move beyond individual transaction columns and examine the

dataset from an operational perspective.

5. Methodology

The analysis followed a structured workflow:

1. Data loading and validation — the dataset was loaded into Python using Pandas, timestamps were parsed, and the dataset was checked for missing and duplicate records.
2. Data preparation — temporal fields and operational features were created.
3. Exploratory data analysis — activity was aggregated by hour, season, day type, day of week, and day-hour combinations.
4. Operational analysis — congestion and idle-capacity rates were examined by hour and other categorical groups.

5. Visualization — Plotly charts and Streamlit KPI components were used to communicate results.

6. Dashboard development — filters for year, season, day type, and date range were incorporated so users could explore different subsets of the data.

7. Deployment — the application was deployed using Streamlit Community Cloud and the source project was maintained in GitHub.

6. Results and Findings

The analysis produced several notable findings from the project dataset. The overall mean activity was 98.48, while the overall mean OLI was approximately 0.0067. The calculated overall congestion rate was approximately 5.01% and the idle rate was approximately 8.34%.

The busiest hour identified by the analysis was 12:00, and the Afternoon time band showed the highest overall activity. Weekend activity was higher than weekday activity in the aggregated comparison, with mean values of approximately 9.81 and 7.76 respectively for the project's day-type metric. Summer was identified as the highest activity season, while Off-Season showed the highest idle-capacity level.

Hourly operational analysis indicated that congestion was concentrated mainly around the late-morning to afternoon period. The highest hourly congestion rate in the analyzed summary occurred at 11:00, at approximately

14.55%, followed by 12:00 at approximately 14.04% and 13:00 at approximately 12.86%. Idle-capacity levels were

substantially higher during low-activity hours, particularly overnight.

The day-hour analysis also showed strong weekend activity around midday. Saturday and Sunday both recorded their

strongest activity around approximately 12:00–14:00, indicating that weekend midday periods deserve particular attention for capacity and resource planning.

7. Dashboard Development

An interactive Streamlit dashboard was developed as the final analytical interface. The dashboard is organized into four major areas: Overview, Activity Analysis, Operations, and Data & Recommendations.

The Overview section presents KPIs including Average Activity, Congestion Rate, Idle Rate, Peak Strain, Busiest Hour, Busiest Season, Busiest Day Type, and Records Analyzed. Activity Analysis provides hourly, seasonal, weekday/weekend, day-of-week, and day-hour visualizations. Operations focuses on congestion versus idle-capacity

patterns and operational hotspots. The Data & Recommendations section provides dataset summaries, high-activity

periods, filtered-data viewing, download options, data-quality checks, and recommendations.

Interactive filters allow users to select year, season, day type, and a date range. This makes the dashboard useful

not only as a static report but also as an exploratory tool for operational analysis.

8. Operational Recommendations

Based on the observed patterns, several practical recommendations can be derived:

- Monitor late-morning and midday periods closely because congestion is concentrated during these hours.
- Consider aligning staffing and operational resources with the stronger weekend and summer demand patterns.
- Investigate overnight and other low-activity periods for opportunities to improve resource utilization.
- Use the dashboard's time and seasonal filters to support periodic operational reviews.
- Track congestion and idle-capacity indicators together rather than treating either measure in isolation.
- Continue collecting timestamped activity data so that future analyses can evaluate changes in demand patterns over time.

9. Limitations

The findings are based on the available project dataset and its recorded variables. The analysis does not directly model vessel capacity, route-level characteristics, weather, ticket price, special events, or other external factors unless such information is present in the underlying data. Therefore, the identified patterns should be interpreted as analytical indicators rather than causal explanations.

The project also uses a fixed Peak Strain summary value of 585 minutes in the dashboard. This value is retained as

a project-level reported metric rather than presented as a newly calculated dynamic measure. Future versions could

implement a formally validated peak-strain methodology using clearly defined operational thresholds.

10. Conclusion

This project demonstrates a complete data analytics workflow for ferry operational data, from data preparation and feature engineering to exploratory analysis, visualization, dashboard development, and deployment. The analysis identified meaningful differences across hours, seasons, and day types and highlighted periods associated

with congestion and idle capacity.

The Streamlit dashboard converts these findings into an accessible decision-support interface. The project also demonstrates practical use of Python, Pandas, Plotly, Jupyter Notebook, GitHub, and Streamlit in an applied data analytics setting. Overall, the solution provides a foundation for monitoring ferry activity and supporting data-driven capacity and operational planning.

11. References

1. Python Software Foundation. Python Documentation.
2. Pandas Development Team. Pandas Documentation.
3. Plotly Technologies Inc. Plotly Python Graphing Library Documentation.
4. Streamlit Inc. Streamlit Documentation.

5. GitHub, Inc. GitHub Documentation.

These references describe the software technologies used to implement the analytical workflow and dashboard.